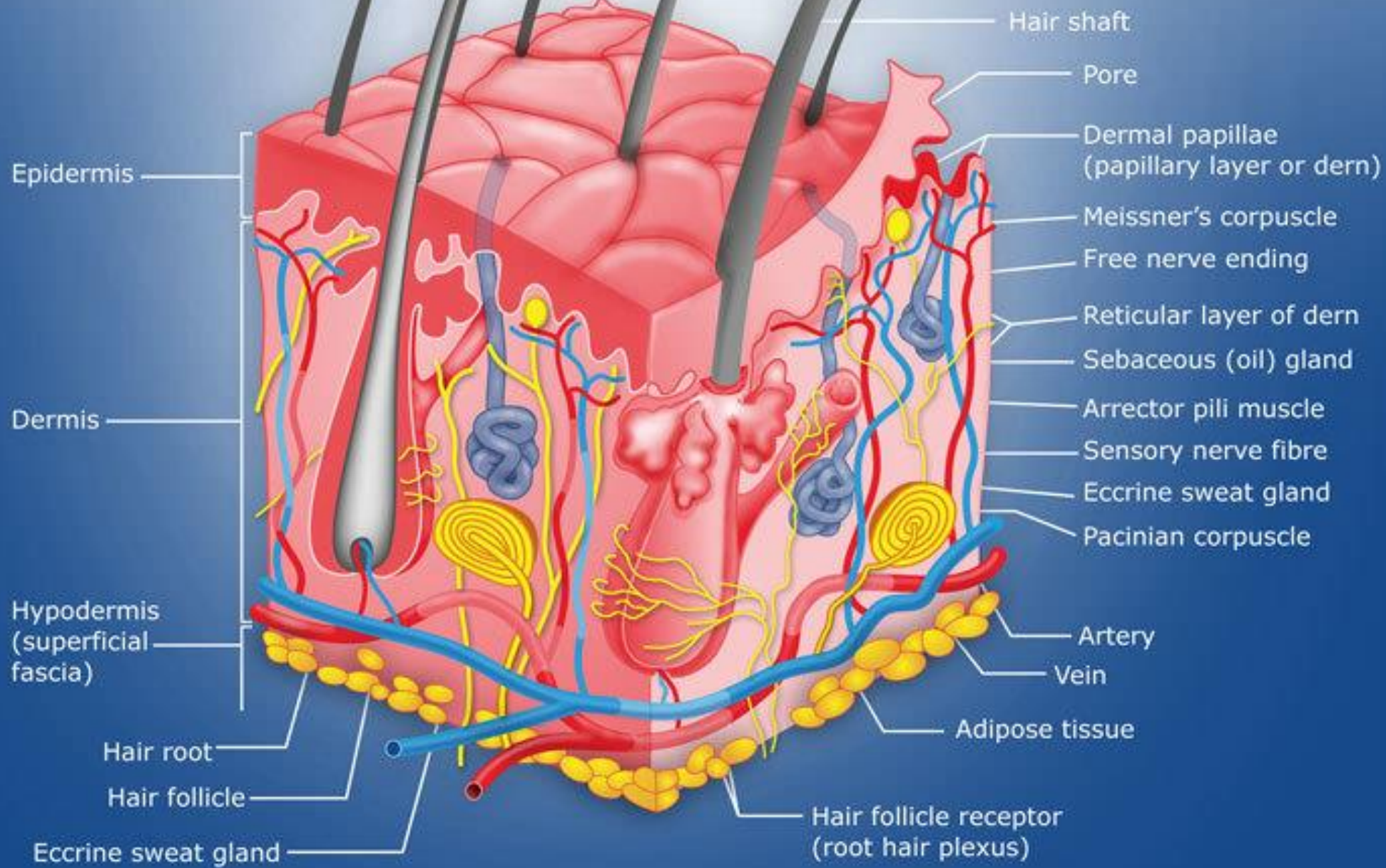


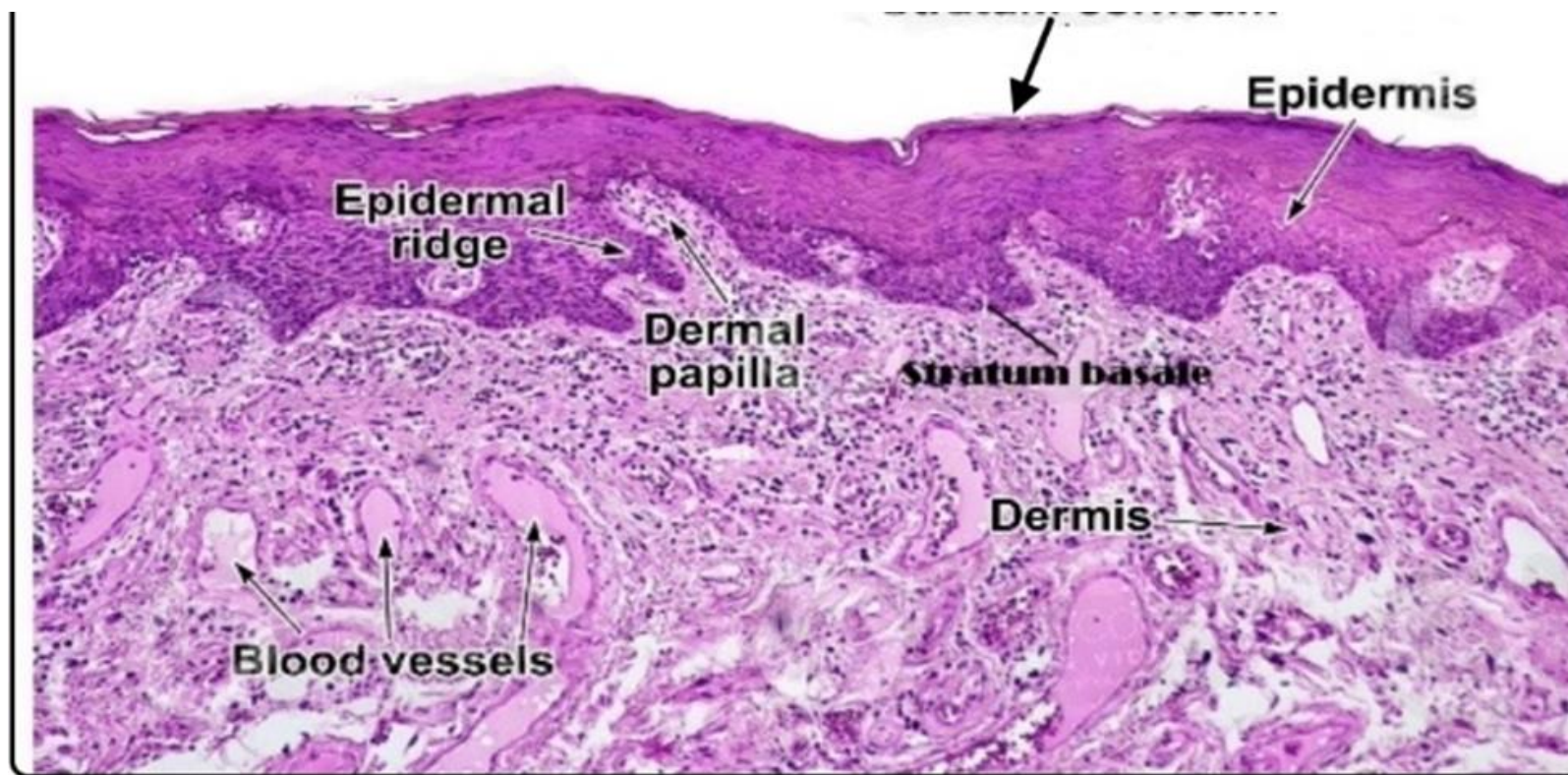
Histology of Human Organs

Human Skin

Human Skin

- The outer covering of the body
- The largest organ of the integumentary system
- **Functions:** Protection, Sensation, Heat regulation, Control of evaporation, Excretion, Aesthetics and communication, Storage and synthesis, Absorption, and Water Resistance
- **Types of skin:**
 1. Hairy skin (everywhere on body)
 2. Glabrous skin (lips, palms of hands, soles of feet)
- **The two primary layers of skin:**
 1. Epidermis
 2. Dermis

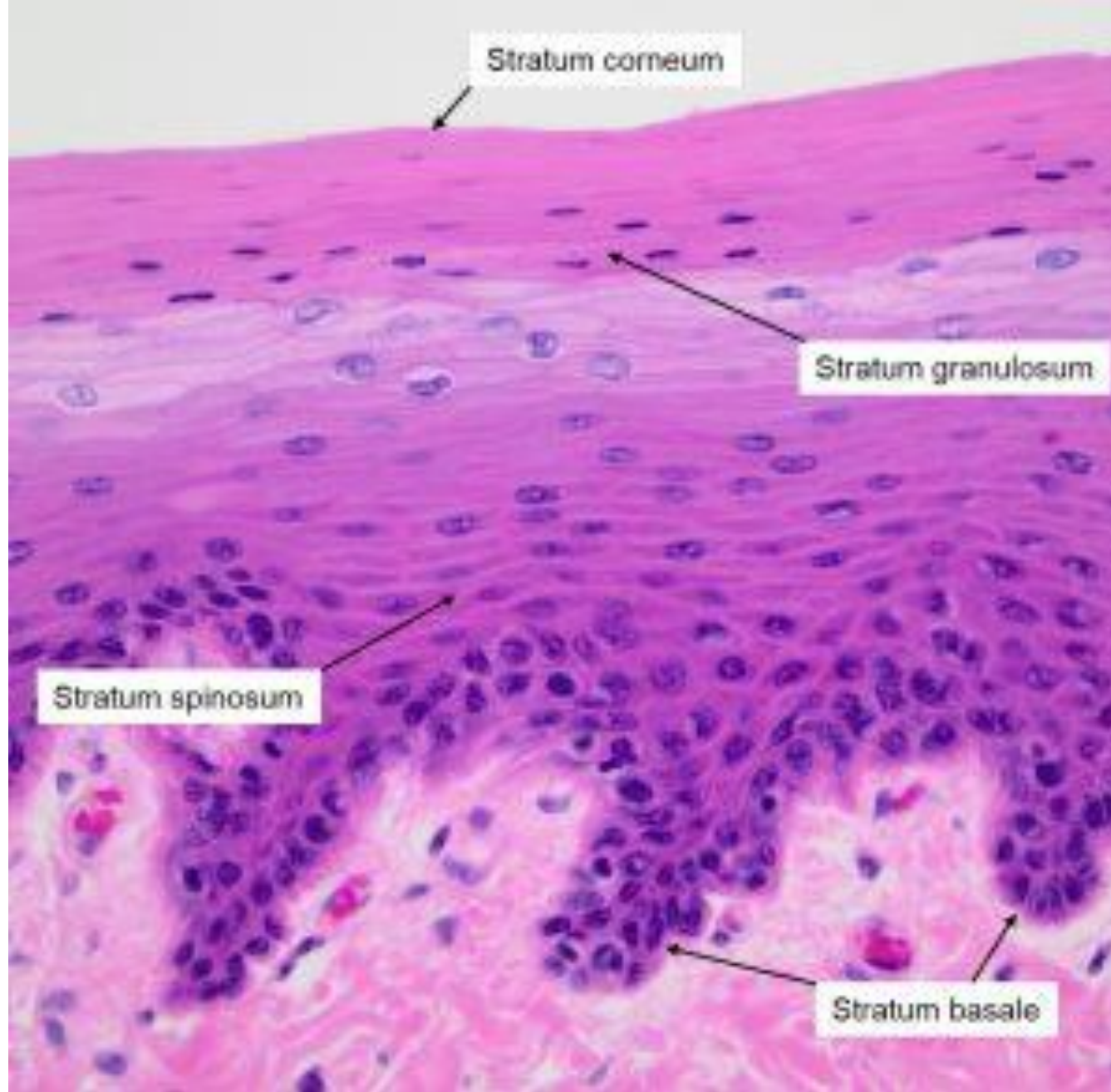




Cross section of Human's skin

1. Epidermis

- The outermost layer of the skin
- **Consists of:** stratified squamous epithelium
- **Function:** Natural barrier to infection and Waterproof
- Avascular (no blood vessels), so the deepest cells are nourished
 - ✓ Mostly by diffused O₂ from the surrounding air
 - ✓ By blood capillaries extending to the upper layers of the dermis
- **Can be divided into 5 sub-layers:**
 - ✓ Outermost strata: Stratum corneum
 - ✓ Stratum granulosum
 - ✓ Stratum spinosum
 - ✓ and the innermost: Stratum basale



Epidermis layers under microscope

keratinization

- The Basale layer divide by mitosis and the daughter cells move up
- The moved cells became isolated from their blood source
- Then the cells lose their cytoplasm and die
- When the cells reach the Stratum Corneum, it filled with the protein **keratin**
-
- Keratinized layer of skin is a natural barrier that is responsible for:
 - ✓ Keeping water in the body
 - ✓ keeping other harmful chemicals and pathogens out

2. Dermis

- The layer of skin beneath the epidermis
- Consists of connective tissue, tightly connected to the epidermis by basal lamina (basement membrane)
- Dermis layer Contains:
 - ✓ Nerve endings that provide the sense of touch and heat
 - ✓ Blood vessels: provide nourishment and waste removal from its own cells as well as from the Stratum basale of the epidermis
 - ✓ Sweat glands: secrete sweat
 - ✓ Sebaceous glands: secrete sebum (oily material) to lubricate the skin
 - ✓ Hair follicles, sebaceous glands, apocrine glands, lymphatic vessels

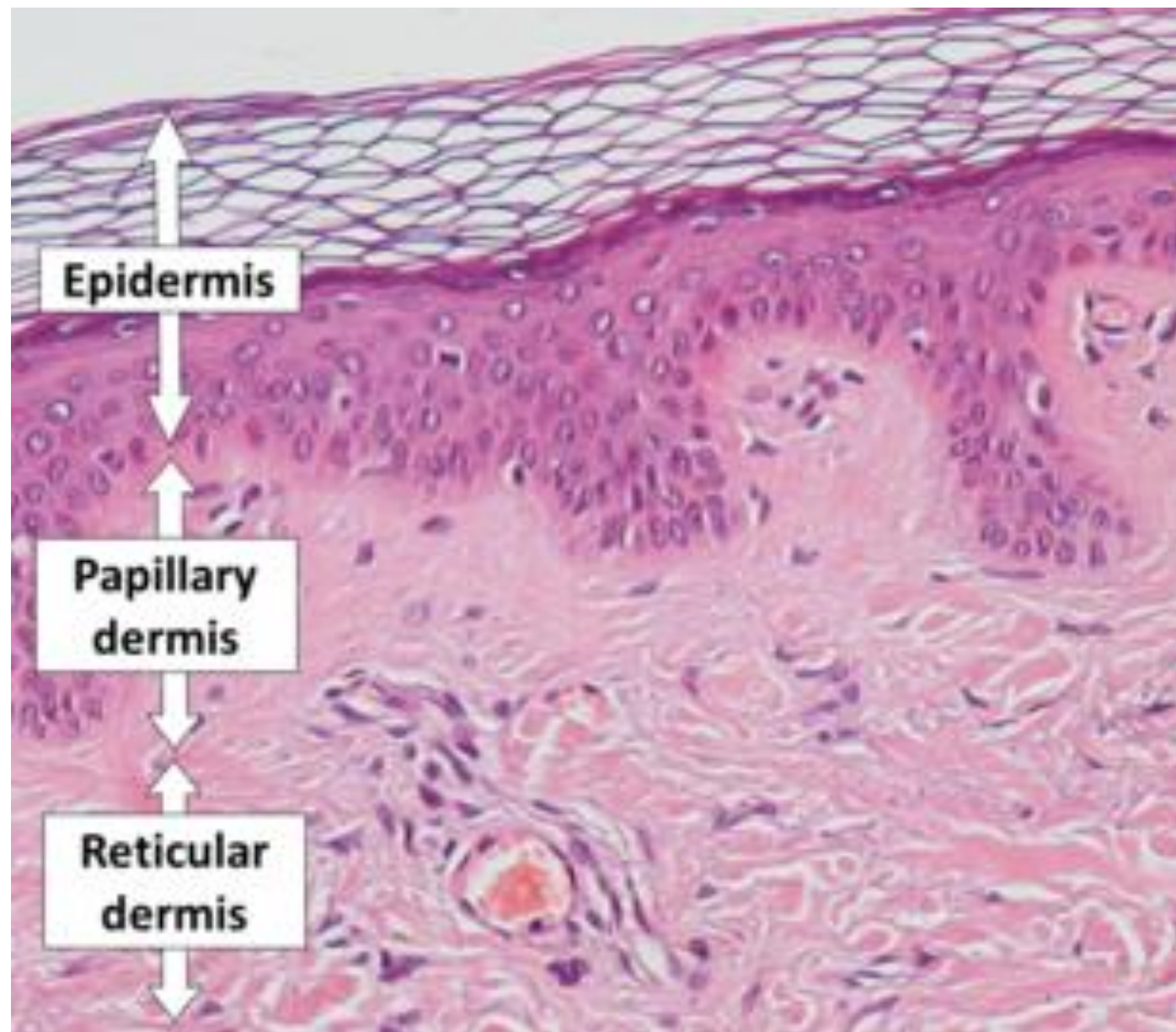
- **The dermis is structurally divided into 2 areas:**

1. Papillary region

- ✓ Finger-like projections
 - ✓ **Consists of areolar (loose) connective tissue**
 - ✓ **Functions:** provides dermis with a bumpy surface that strengthens the connection between Dermis and Epidermis
- Reticular region

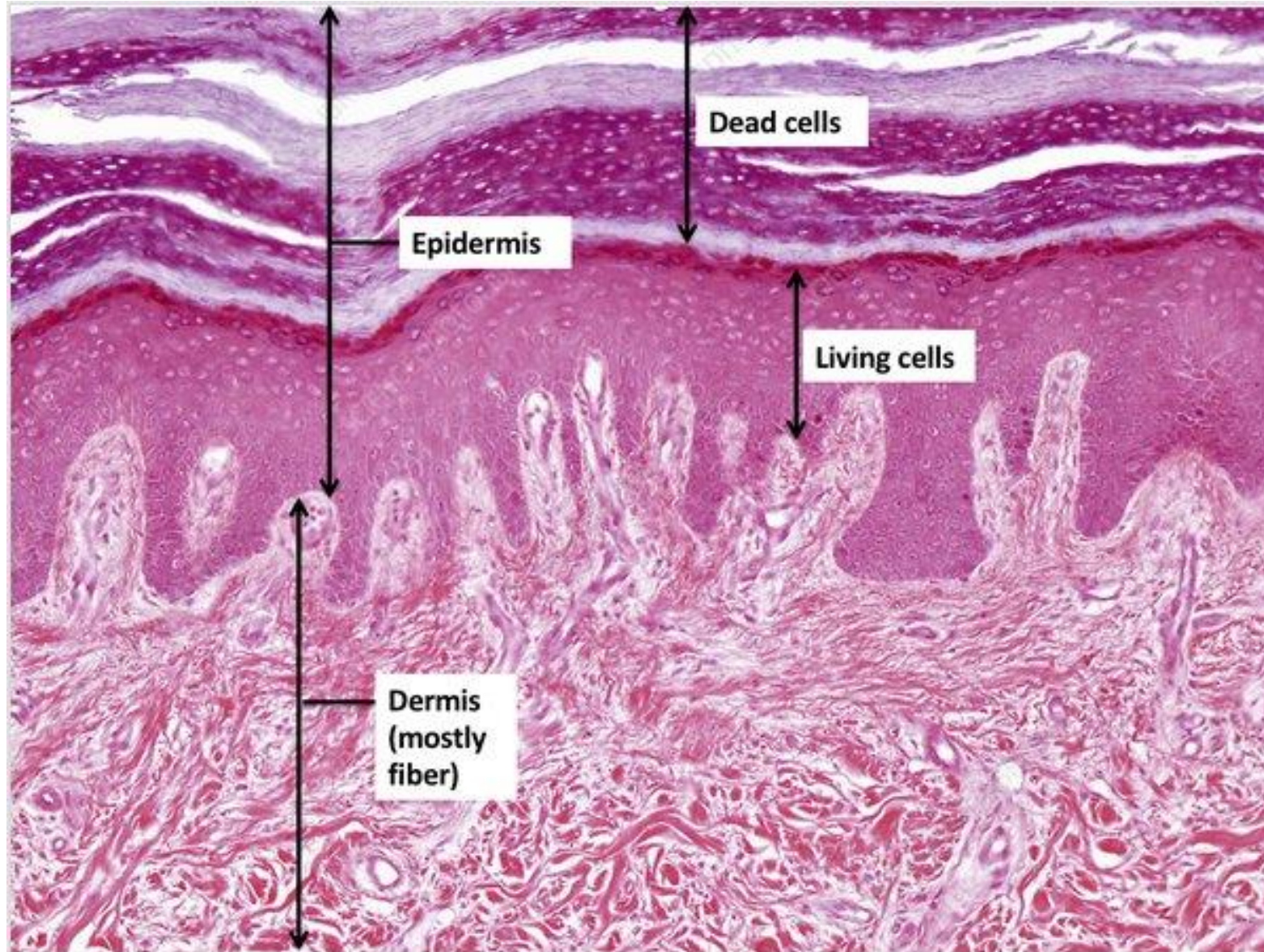
2. Reticular region

- ✓ **Consists of dense irregular connective tissue** that contains collagenous, elastic, and reticular fibers
- ✓ **Function:** The fiber gives the dermis its strength, elasticity, and extensibility
- ✓ Contains sebaceous glands, which produce sebum (a naturally healthy skin lubricant)



Dermis layers under microscope

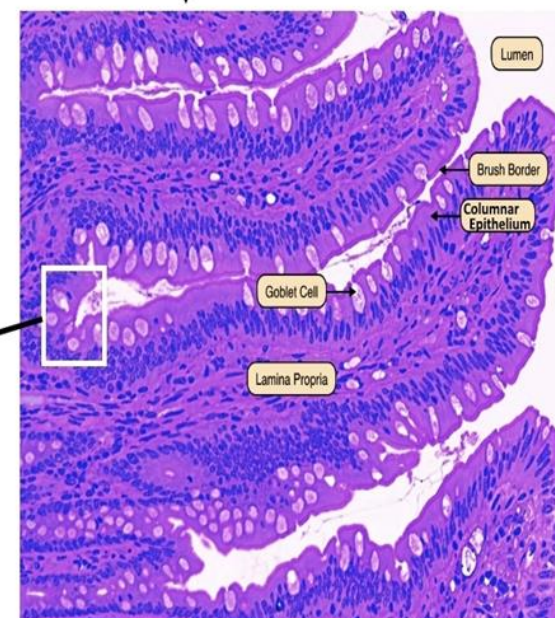
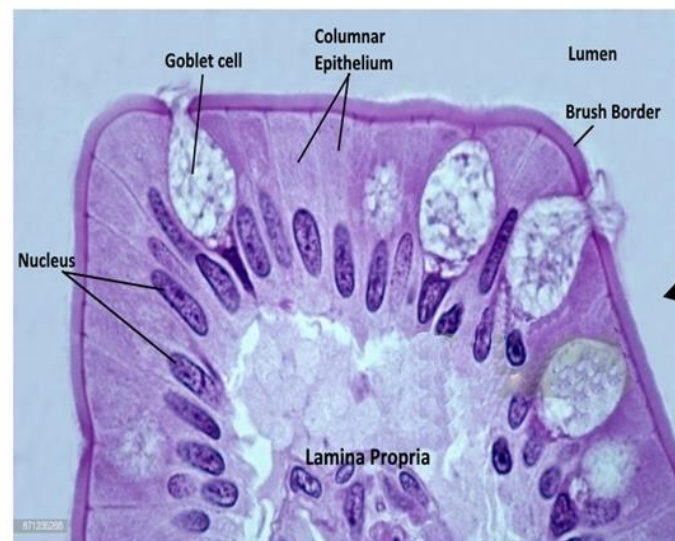
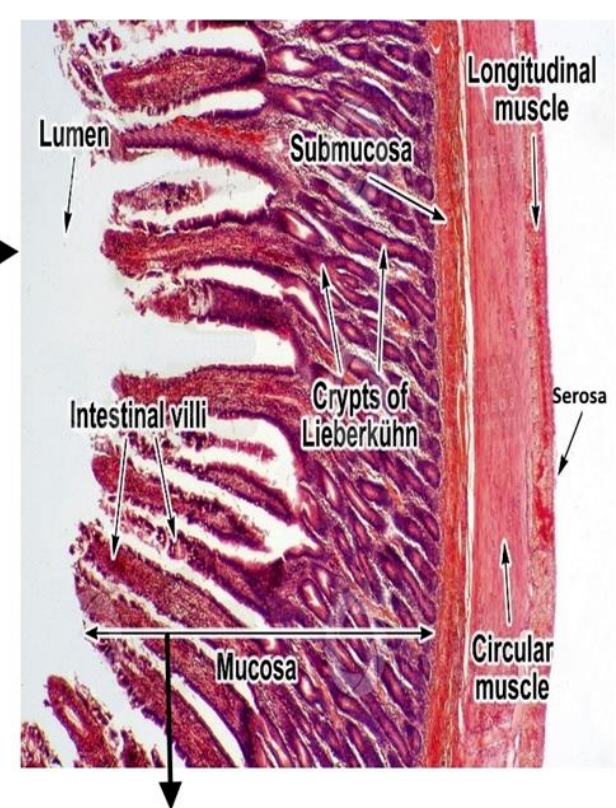
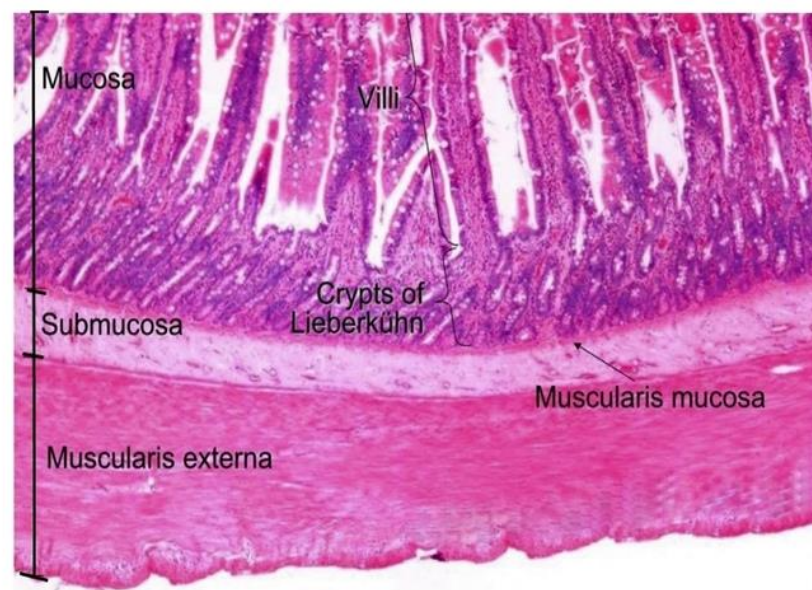
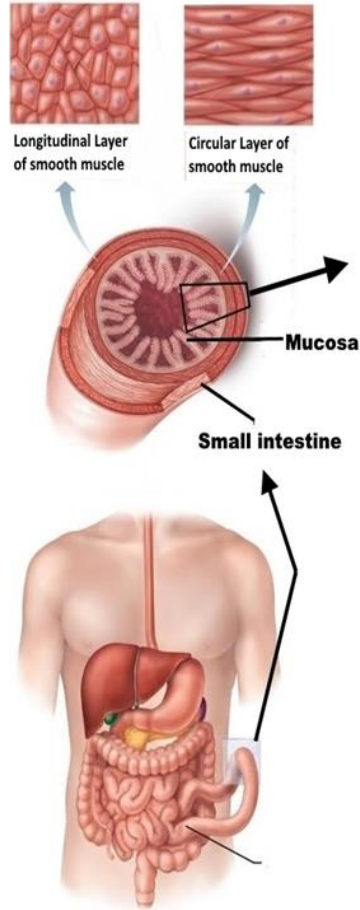
To Sum Up



Human Intestine

Human Intestine

- **Function:** digests food to enable the nutrients released from food to enter the bloodstream
- Together the intestines take up most of the space within the abdominal body cavity and are folded over many times
- **Consists of 2 major subdivisions:**
 1. **Small intestines**
 - ✓ Much smaller in diameter but longer and larger than the large intestine
 - ✓ Digestion of food and absorption of nutrients
 2. **Large intestines**
 - ✓ Larger in diameter and shorter length
 - ✓ Absorption of water



- **The intestinal wall is divided into 4 layers (seen in both the jejunum and duodenum):**
- Innermost (mucosa, submucosa & muscularis) & Outermost (serosa)

1. Mucosa

- Contains villi, which are finger-like projections lined with **simple columnar epithelial tissue** (have an ovoid nucleus at their base) and goblet cells
- The epithelial cells of the mucosa absorb the nutrients and water from digested food and transfer these substances to blood in nearby capillaries
- Goblet cells secrete mucus to prevent friction from food passing thru the lumen
- Basement membrane: Support the epithelium layer and is made of reticular fibers embedded in an amorphous protein-polysaccharide ground substance

2. Submucosa

- **Composed of loose connective tissue**

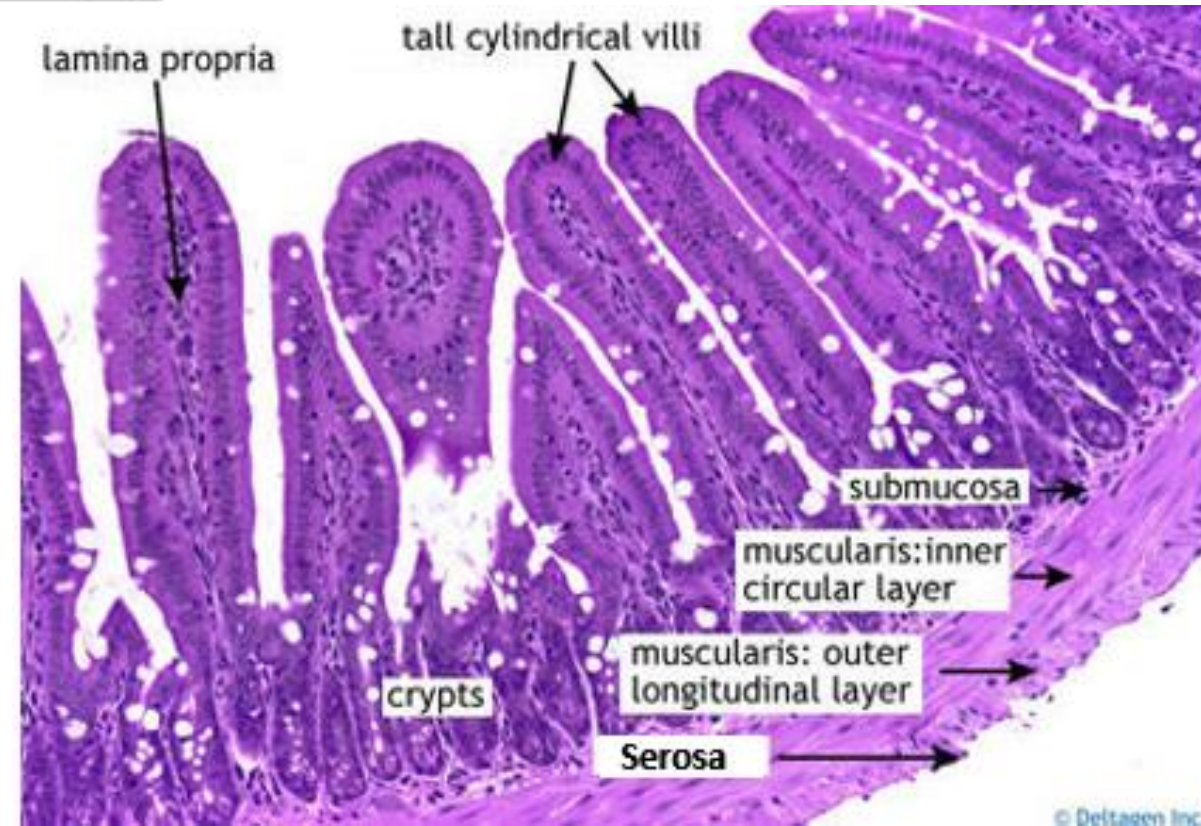
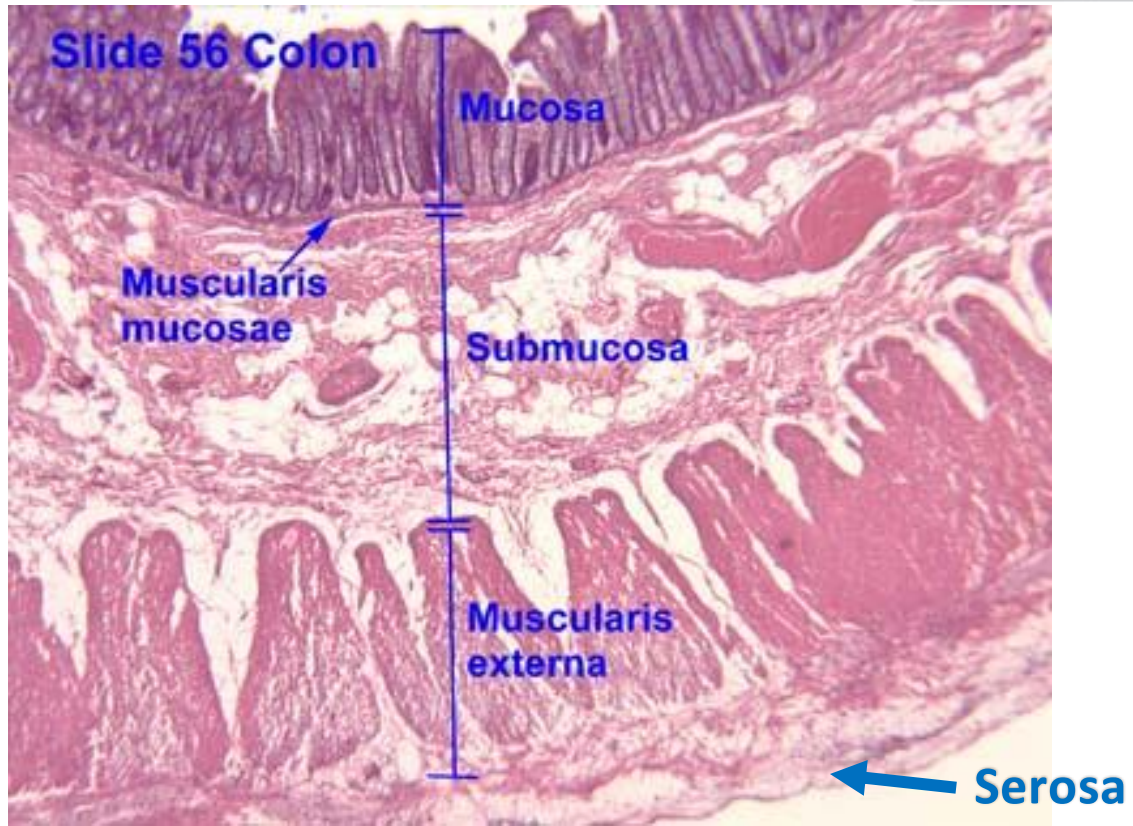
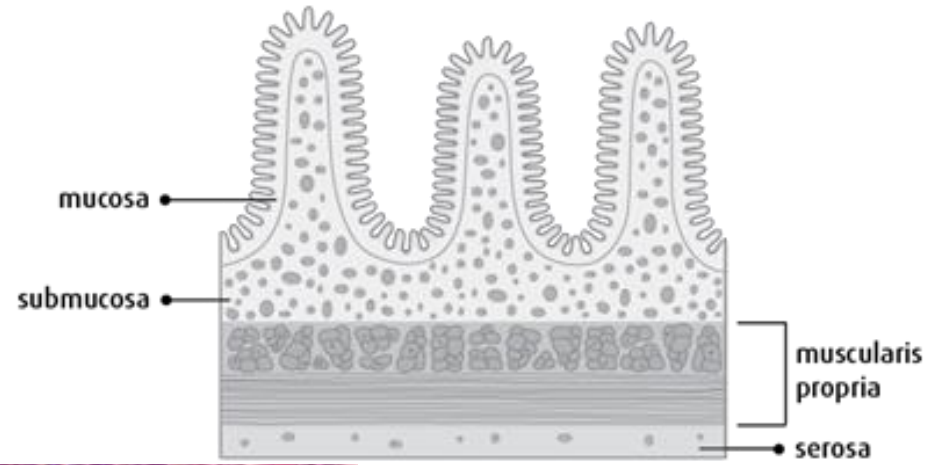
3. Muscularis

- Contains **inner circular and outer longitudinal layers of smooth muscle fibers** to move the intestines
- The inner and outer layers are separated by reticular and collagenous connective tissue containing nerve fibers and parasympathetic ganglion cells

4. Serosa

- The outermost consists of loose connective tissue and mesothelium (simple squamous) and loose connective tissue

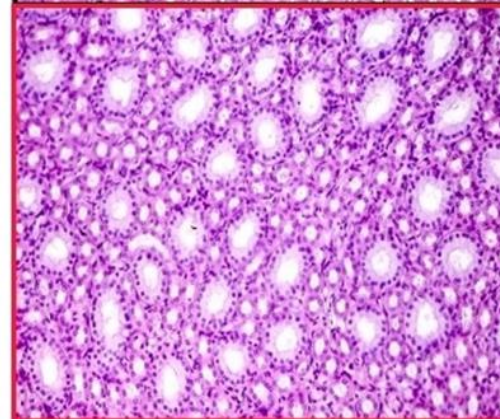
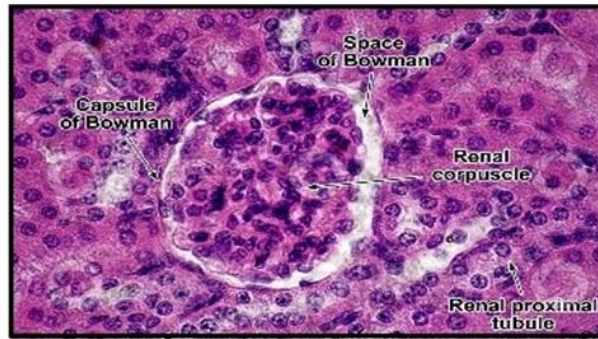
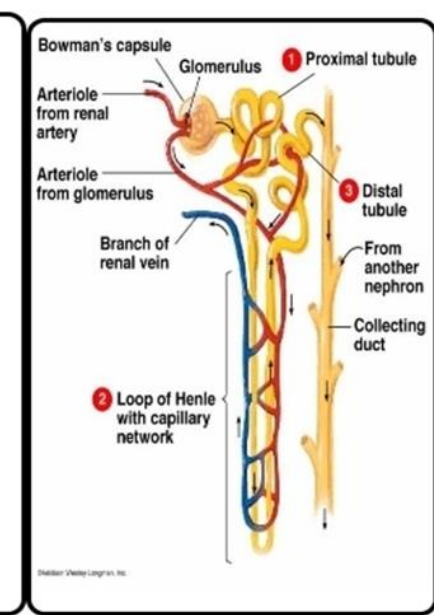
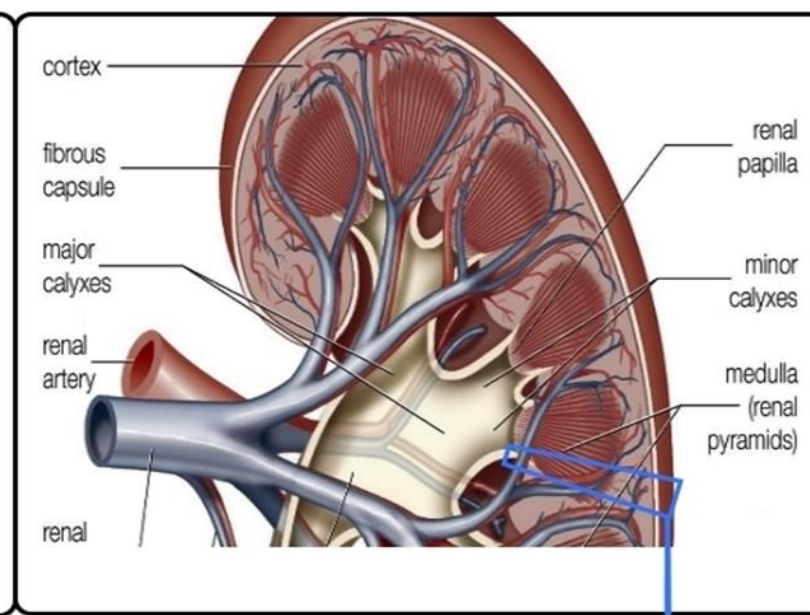
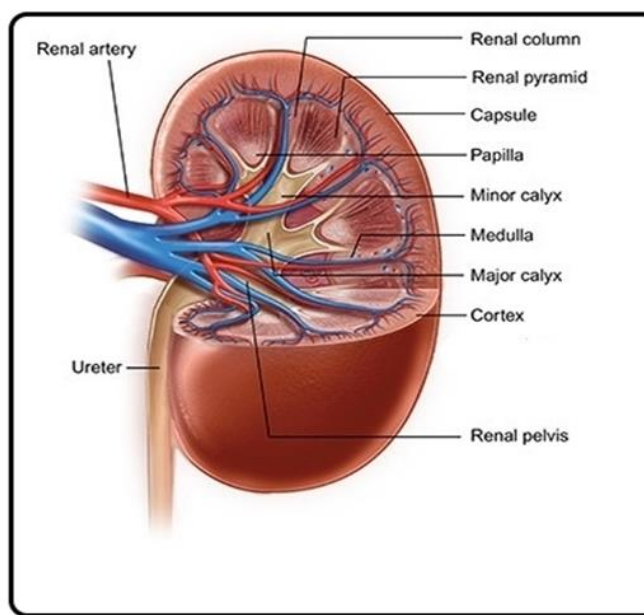
Layers of the Small Intestine



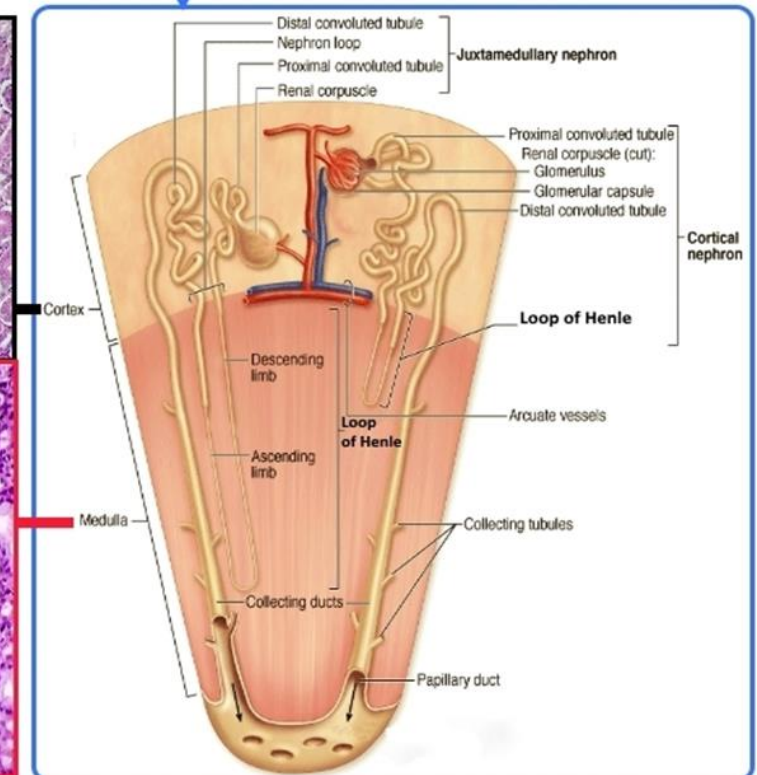
Human Kidney

Human Kidney

- Two bean-shaped organs on both sides of the body, and underneath the diaphragm
- Resting on top of each kidney is an **adrenal gland**
- **Kidney Functions:**
 - ✓ Remove waste products from the body.
 - ✓ Remove drugs from the body.
 - ✓ Balance the body's fluids.
 - ✓ Release hormones that regulate blood pressure
- **Kidney made up from:**
 1. The **cortex** is the outer layer and contains the kidney's filtering structures
 2. Renal cortex
 3. Renal capsule

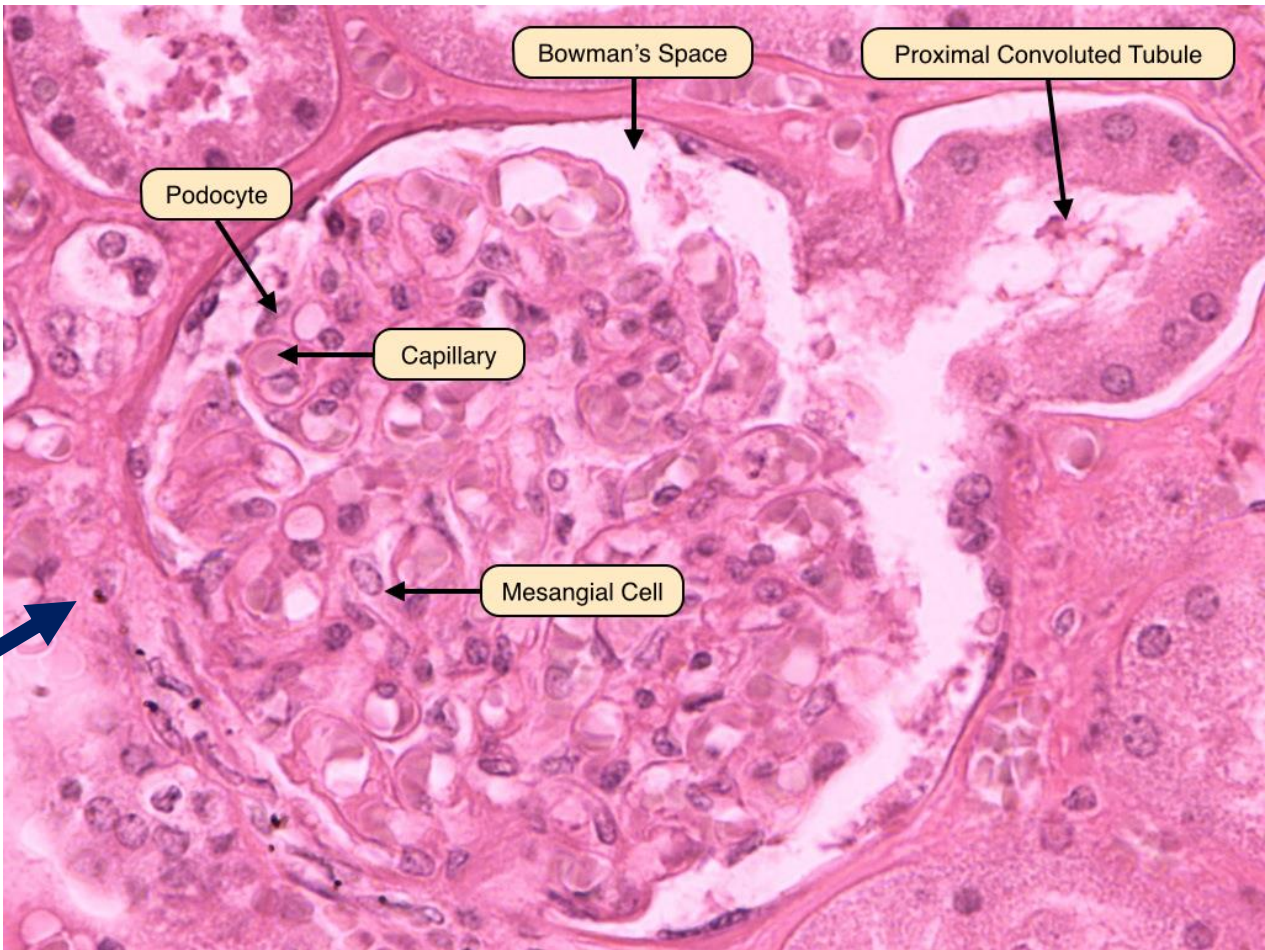
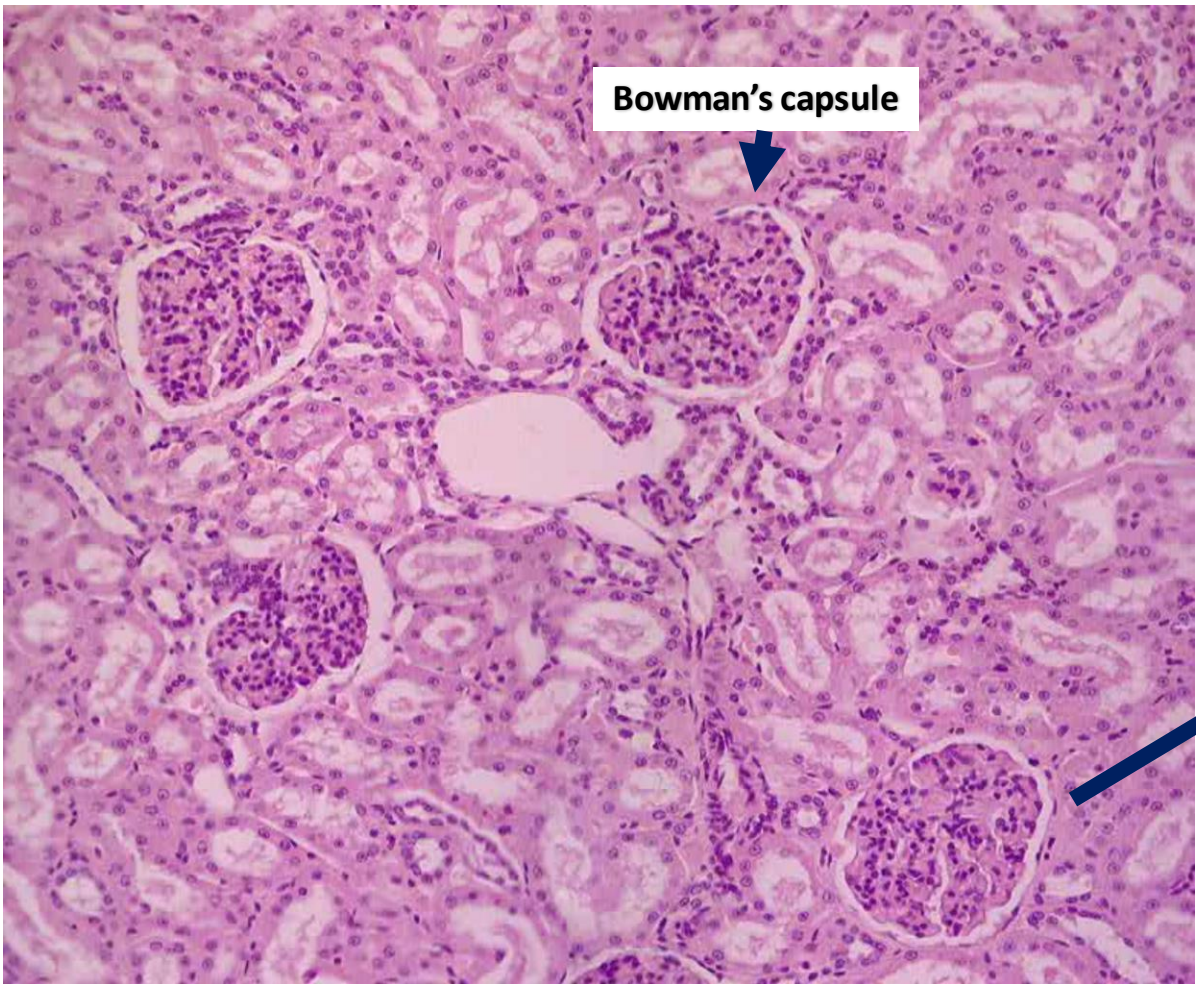


Human's Kidney



1. Renal Cortex

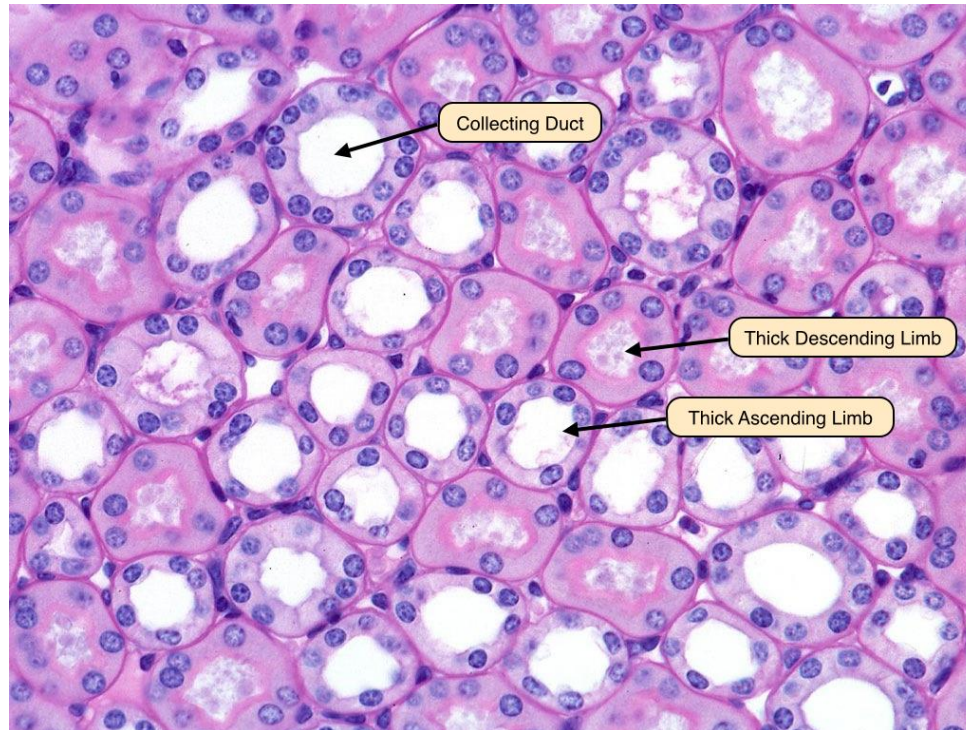
- The renal cortex is the outer layer of the kidney.
- The cortex surrounds the inside of the kidney, which is called the medulla.
- The renal cortex is where the nephrons (blood-filtering units) begin.
- The renal cortex itself is covered by the **renal capsule**, which is a layer of tougher protective tissue.
- The renal capsule consists of:
 - ✓ A layer of myofibroblasts called the inner layer of the capsule
 - ✓ Loose connective tissues are called the outer layer of the capsule.



Renal Cortex

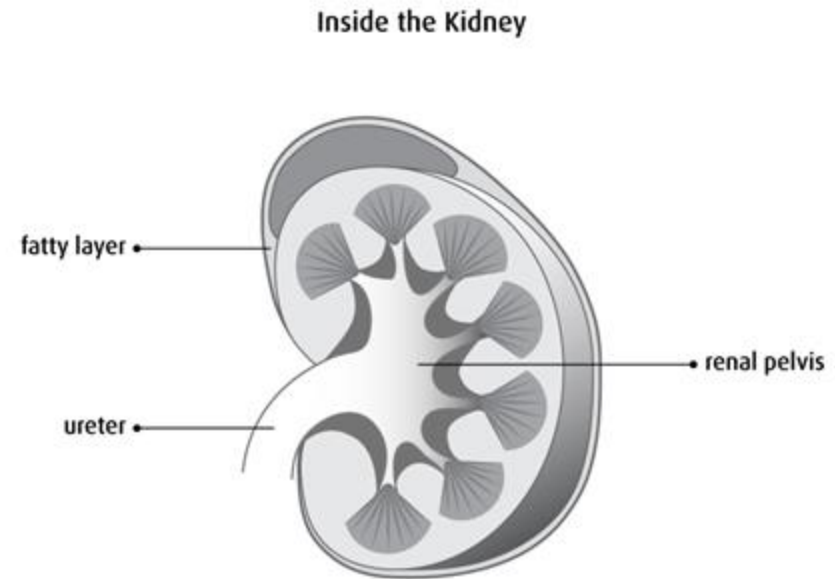
2. Renal Medulla

- The renal medulla is the inner part of the kidney.
- Function: the medulla helps regulate the concentration of urine by filtering out water, salts, and acid.



3. Renal Pelvis

- The area at the center of the kidney.
- Urine collects here and is funneled into the ureter, the tube that connects the kidney to the bladder.



Nephron

- The nephron is the functional unit of the kidney.
- There are about two million nephrons in each kidney.
- Nephrons begin in the cortex as corpuscles; become tubules and dip down to the medulla; then return to the cortex before draining into the collecting duct.
- The collecting ducts then descend towards the renal pelvis and empty urine into the ureter.
- The components of a single nephron include:
 - ✓ renal corpuscle
 - ✓ proximal convoluted tubule
 - ✓ loop of Henle
 - ✓ distal convoluted tubule

Human Liver

Human Liver

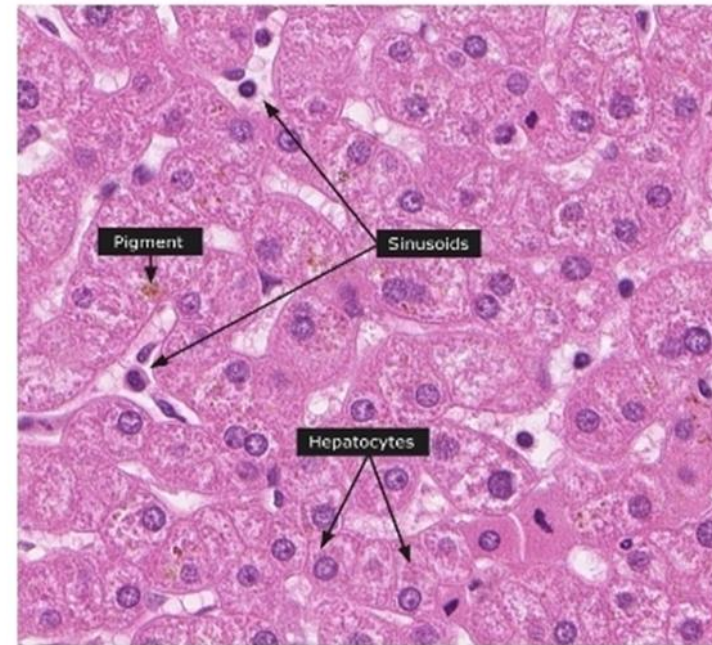
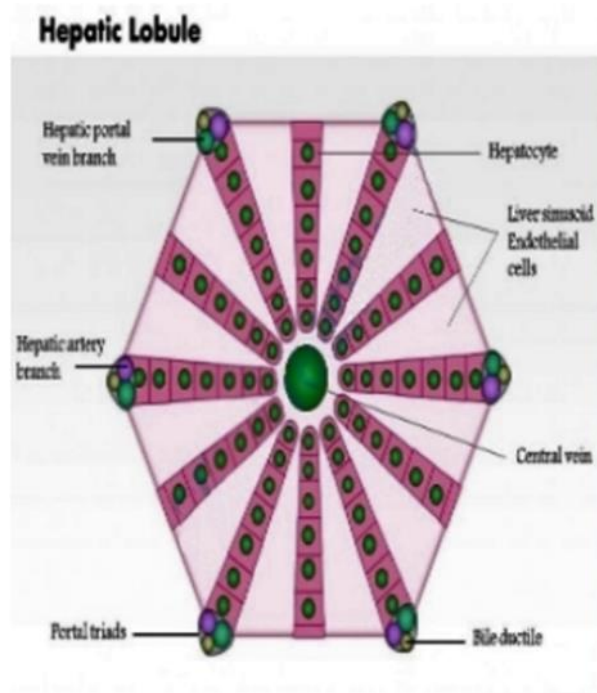
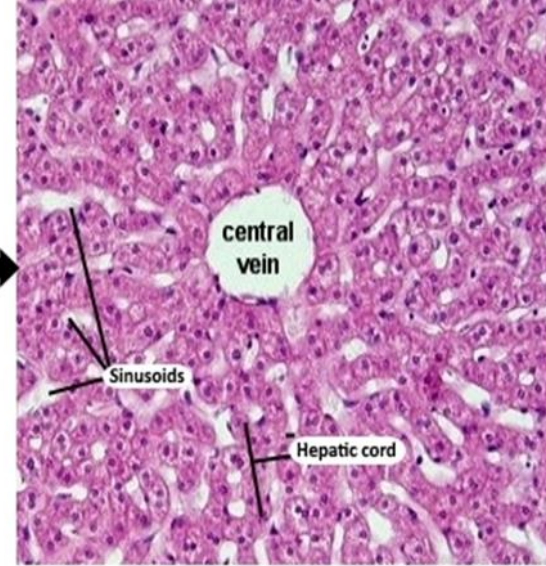
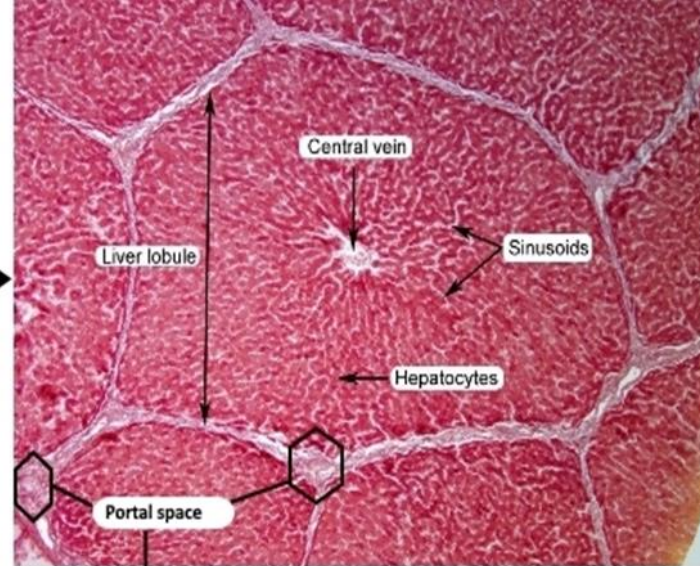
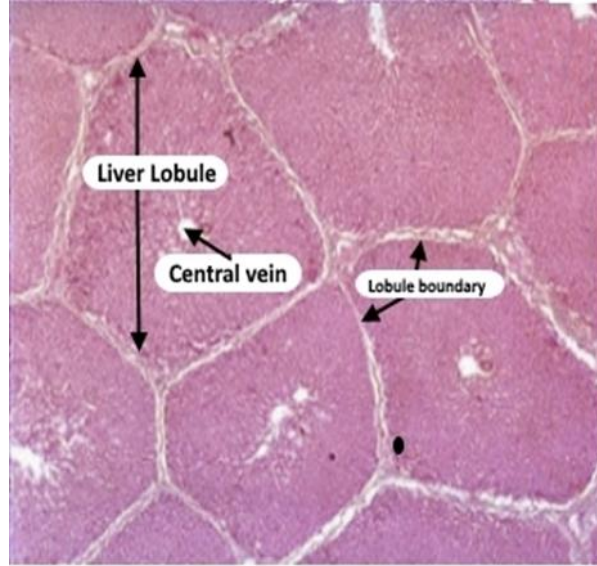
- Largest gland in body, Divided into 4 lobes
- It has both:
 - ✓ Exocrine gland: secretes bile into duodenum via system of bile ducts
 - ✓ Endocrine: synthesizes and releases organic compounds into the bloodstream
- **A cross section of the liver shows that it composed from:**
 1. **Hepatic lobule:** The functional unit of the liver.
 2. **Central vein:** Forms the axis of the hepatic lobule. Receives blood from the hepatic sinusoids and drains into sub lobular vein.
 3. **Liver cells (acini):** Polyhedral cells with a round central nucleus. Arranged in cords and plates radiating from the central vein.
 4. **Sinusoids:** Form system of vascular channels that radiate from the central vein
 5. **Hepatic cords:** Sheets or plates of hepatic cells
 6. **Bile canaliculi:** These are channels between rows of cells within hepatic cords. Bile enter the system of bile ducts and gallbladder and secreted into the duodenum

7. Hepatocyte:

- Making up 70-80% of the liver's cytoplasmic mass.
- Hepatocytes contain large amounts of rough endoplasmic reticulum and free ribosomes.
- Hepatocytes are involved in protein synthesis, protein storage, transformation of carbohydrates, synthesis of cholesterol, bile salts and phospholipids, and detoxification, modification, and excretion of exogenous and endogenous substances.

8. Portal space:

- The space between hepatic lobule. It contains:
 1. A branch of the bile duct
 2. A branch of hepatic Artery
 3. A branch of portal Vein.



Bile Salts

What is bile made of?

- Bile salts composed of water, bile pigment, cholesterol, lecithin, fat, and inorganic salts
- Bile salts are the only part of the bile to be recycled by absorbing it during digestion and returned to the liver for reutilization.

- **What is the function of the bile?**

1. Acceleration the action of pancreatic and intestinal lipases
2. Facilitates the absorption of fats from the intestine

Human Trachea

Human Trachea

- Known as the windpipe
- 4 inches long and less than an inch in diameter for most people
- Begins just under the larynx (voice box)
- Runs down behind the breastbone (sternum) divides into 2 bronchi, one bronchus for each lung
- Trachea widens and lengthens slightly with each breath in, and returning to resting size with each breath out
- **Tracheal wall made up of 4 layers:**
 1. Mucosa
 2. Submucosa
 3. Hyaline cartilage
 4. Adventitia

1. Mucosa

- Consists of **ciliated pseudostratified columnar epithelium** with many goblet cells
- Goblet cells produce sticky mucus to coat the inner lining of the trachea to catch any debris present in inhaled air
- Cilia beat together to push mucus away from lungs Mucus, along with trapped contaminants, is either
 - ✓ expelled through the larynx by coughing
 - ✓ swallowed and digested in the stomach

2. Submucosa:

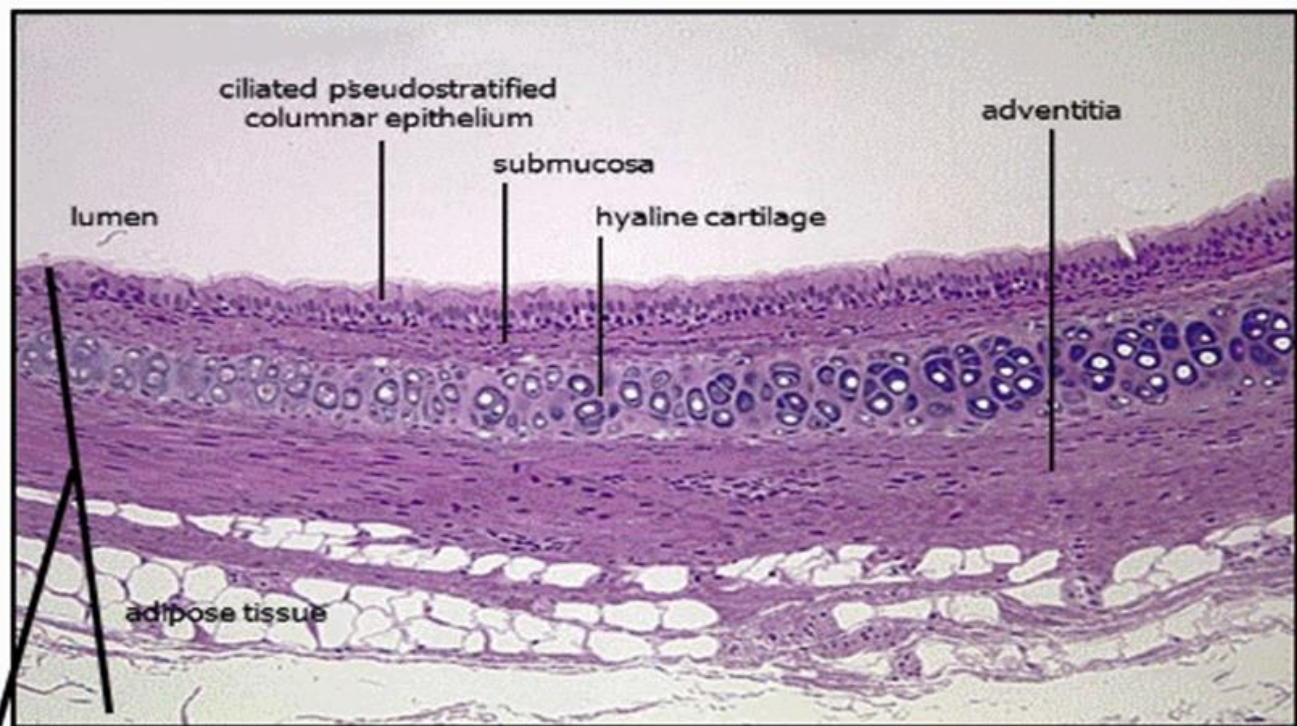
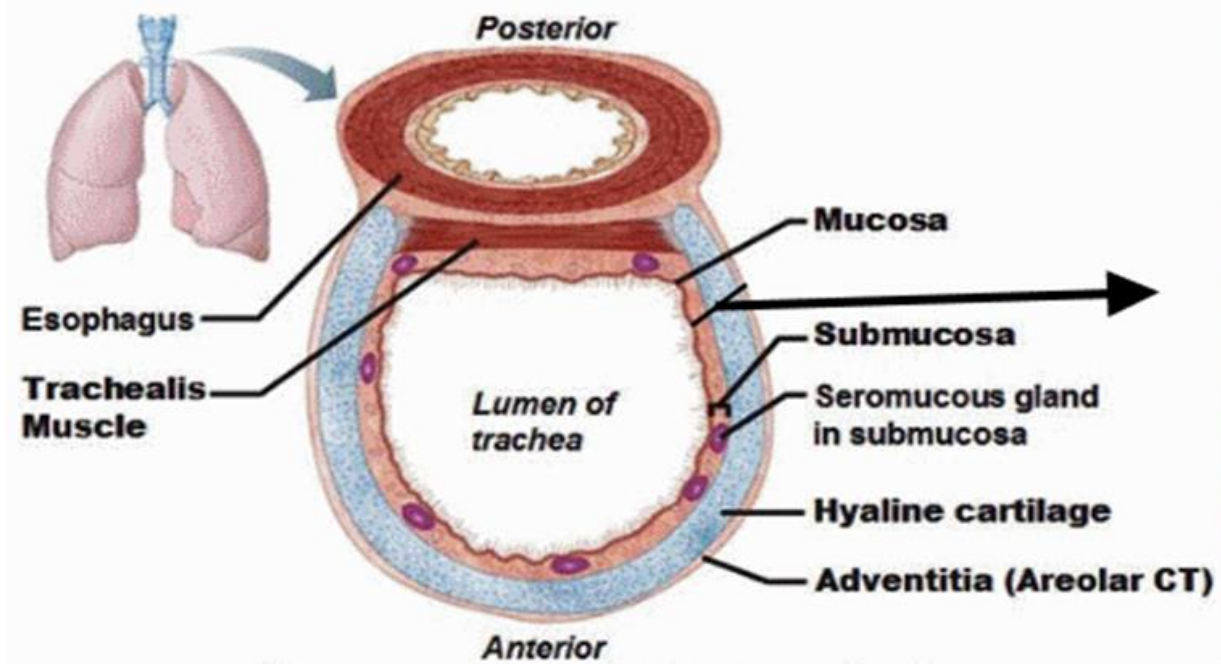
- Made of **areolar connective tissue**
- Many collagen, elastin, and reticular protein fibers give soft support and elasticity to the other layers of tracheal wall
- **Longitudinal smooth muscle fibers** are present in the posterior trachea between the ends of the cartilage rings
- Smooth muscle tissue allows the trachea to adjust its diameter as needed

3. Hyaline cartilage

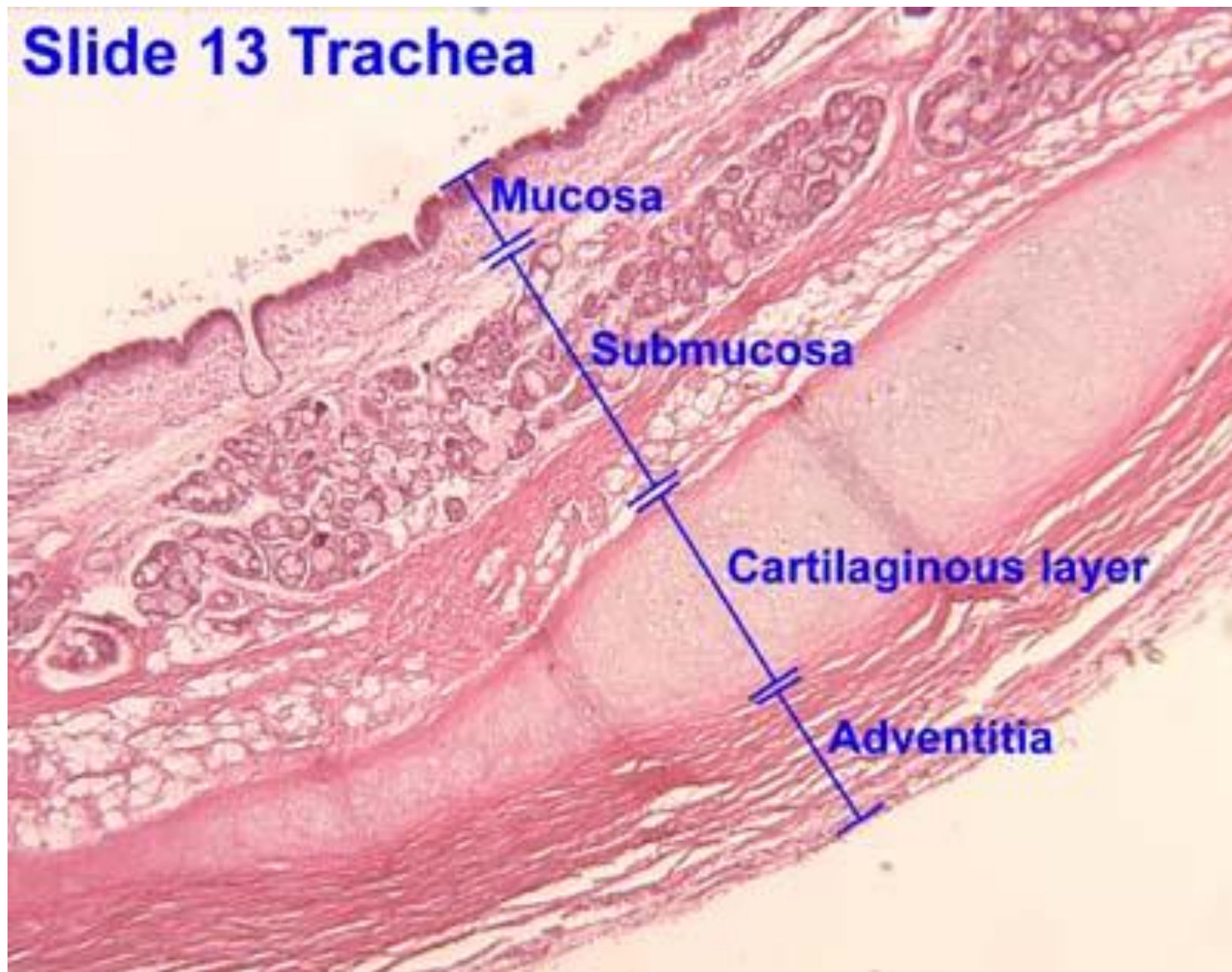
- C-shaped rings that maintain open airway
- Flexible structure that maintains an open airway and is resistant to external stresses
- Contains lacunae where chondrocytes are located

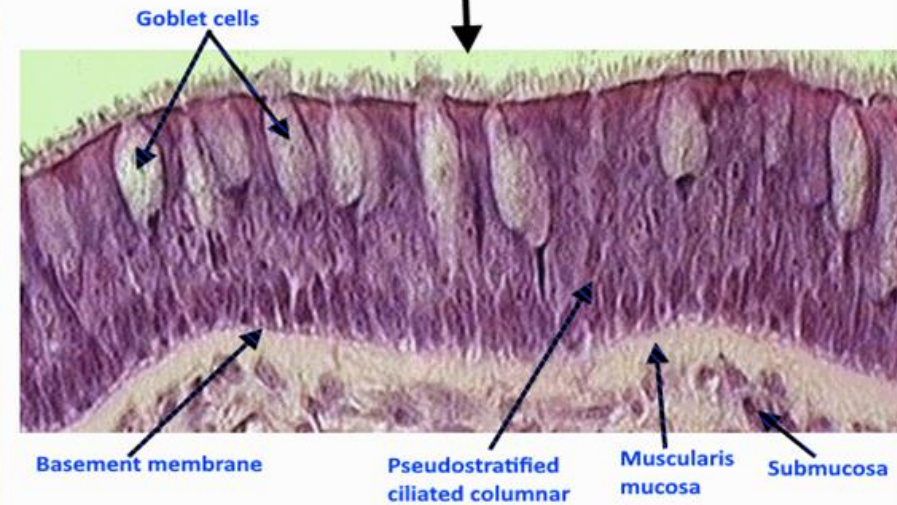
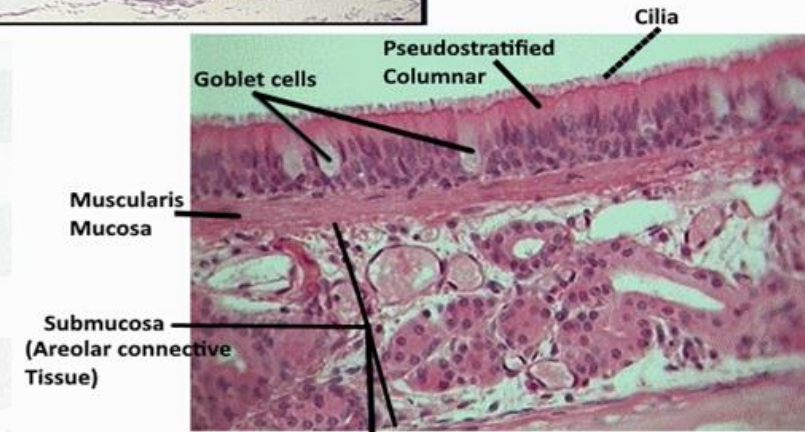
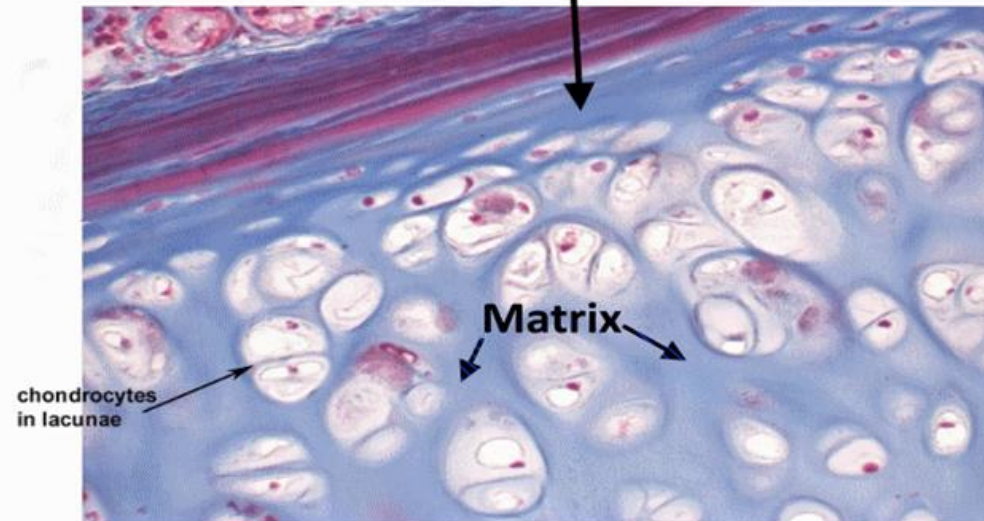
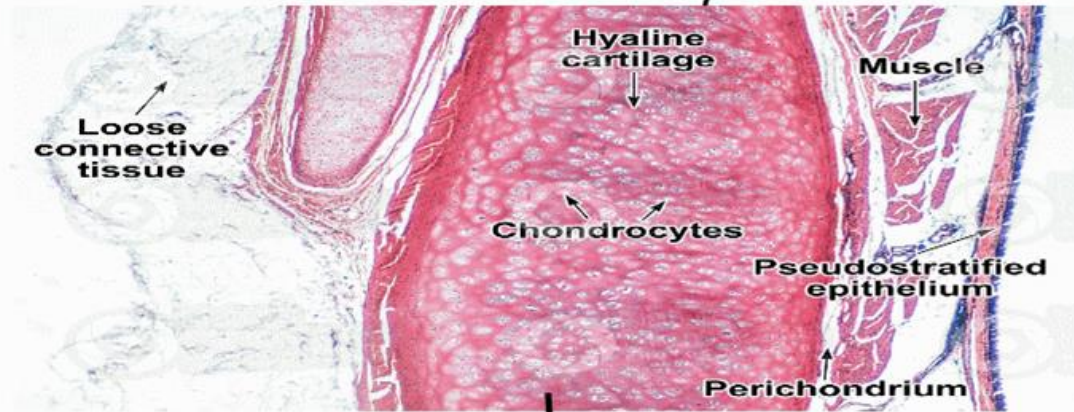
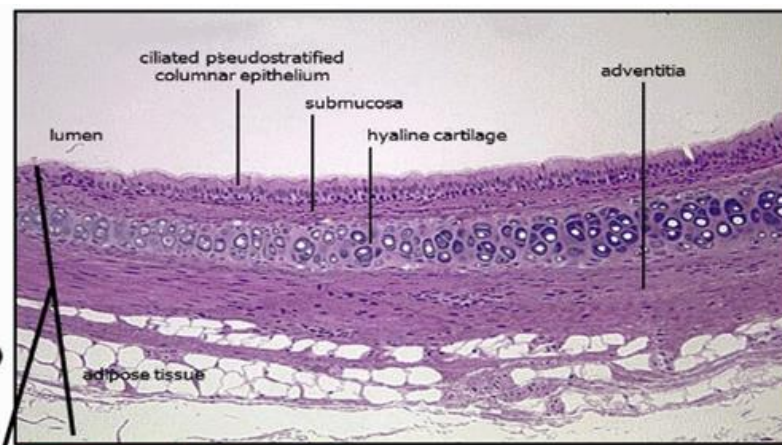
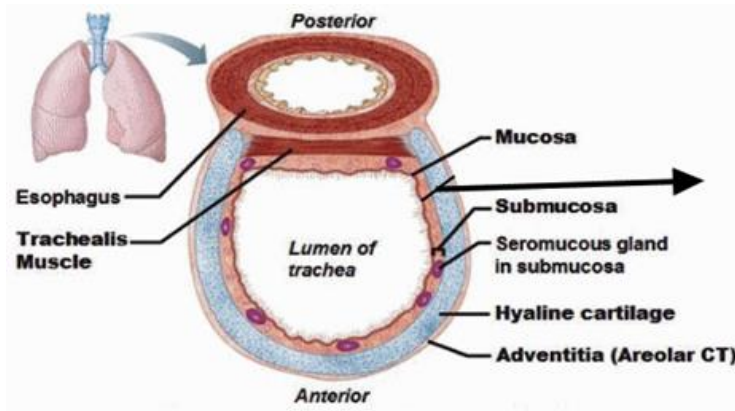
4. Adventitia

- Layer of areolar connective tissue that loosely anchors the trachea to the surrounding soft tissues



Slide 13 Trachea

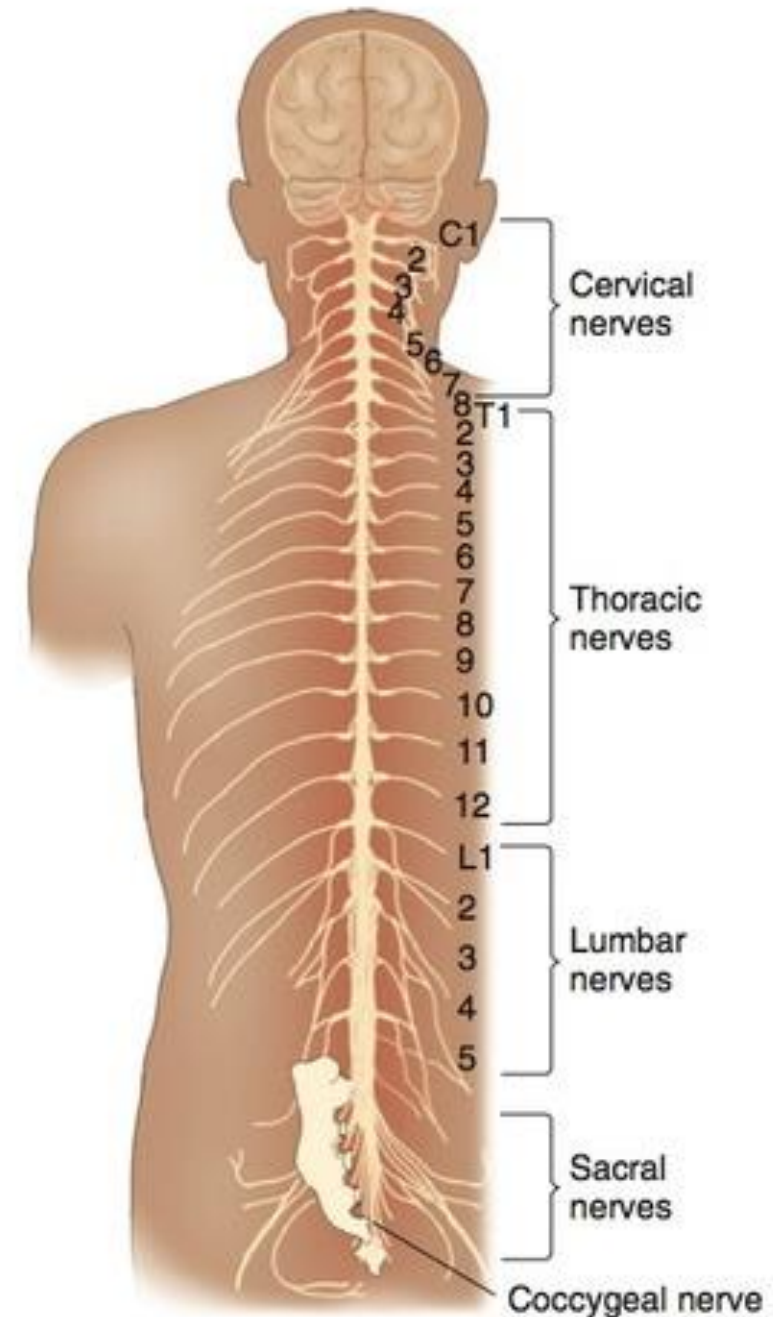




Human Spinal Cord

Human Spinal Cord

- The nervous system divides into two: The central nervous system (CNS) and the Peripheral nervous system
- The central nervous system (CNS) includes: Brain + spinal cord
- Divided into 31 pairs of nerves (62 in total), named according to where they leave the spine (5 groups):
 1. 8 pairs in the cervical region
 2. 12 pairs in the thoracic region
 3. 5 pairs in the lumbar region
 4. 5 pairs in the sacral region
 5. 1 pair in the coccygeal region



- Across section in the spinal cord shows that the spinal cord contains:
 1. White matter
 - ✓ Contains myelinated axons, Travels in tracts to and from the brain
 2. Grey matter: H-shaped,
 - ✓ Contains: Neurons, glial cells, and nonmyelinated axons

- Grey matter contains 3 horns on either side of the spinal cord:
 1. Ventral horn: Contains somatic motor nuclei
 2. Lateral horn: Contains autonomic motor nuclei
 3. Dorsal horn: Divided into somatic sensory portion and visceral sensory portion

